

SHAIK MOHAMMED SAMEER

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PROFESSIONAL SUMMARY

Backend engineering student with hands-on experience building production-grade API gateways, rate limiters, and authentication systems using Node.js, NestJS, TypeScript, and Redis. Proven ability to design distributed, scalable microservices architectures through independent projects and internship experience. Strong foundation in data structures, algorithms, system design, and Agile development practices.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, SQL

Frameworks: Node.js, NestJS, Express.js, React, Next.js

Databases: PostgreSQL, Redis, MongoDB (basic)

Tools: Docker, Git, GitHub, Jest, Postman

Concepts: REST APIs, JWT Auth, Rate Limiting, Microservices, System Design, DSA, CI/CD basics, Observability

EXPERIENCE

Software Development Trainee | NoviTech R&D Pvt. Ltd.

Oct 2024 – Nov 2024

- Developed 8+ RESTful API endpoints and implemented JWT and API-key authentication modules for 2 web applications.
- Integrated frontend and backend systems, reducing data-fetch round trips by refactoring API response structures.
- Managed PostgreSQL database operations including schema design, queries, and basic optimizations.

PROJECTS

Distributed API Gateway & Traffic Management Platform — NestJS, TypeScript, Redis, PostgreSQL, Docker, Next.js

- Architected a microservices-based API gateway routing requests across multiple backend services with sub-100ms overhead.
- Implemented Redis token-bucket rate limiting, handling 500+ concurrent requests and reducing API abuse incidents to zero in load tests.
- Developed authentication middleware supporting API key and JWT validation, securing 100% of inbound traffic.
- Built an observability pipeline tracking latency, error rates, and 10+ custom traffic metrics with structured request logging.
- Created a real-time monitoring dashboard in Next.js visualizing 5+ KPIs including throughput, p95 latency, and error rates.
- Load tested the gateway using k6, sustaining 500 concurrent users with <200ms average response time and zero downtime.

SRBench – Structured Mathematical Reasoning Benchmark for LLMs — Python, GPT, Qwen, DeepSeek, NLP

- Designed and generated 1,245 template-based math problems spanning 8 difficulty categories for LLM evaluation.
- Benchmarked 3 LLMs (GPT, Qwen, DeepSeek) using exact-match scoring; GPT achieved highest accuracy at 78% vs 61% for DeepSeek.
- Engineered prompt templates and evaluation pipelines using Python, ensuring full dataset reproducibility and quality.

Published: <https://shorturl.at/3jqDL>

EDUCATION

B.Tech in Computer Science

Expected: May 2027

SRM Institute of Science and Technology, Chennai | CGPA: 8.10 / 10